Feasibility Report: PII Detection and Redaction in Images

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1. Executive Summary

This feasibility report evaluates the practicality of implementing an automated system for detecting and eliminating or redacting Personally Identifiable Information (PII) from images. With increasing regulatory pressure (e.g., GDPR, DPDP Act, HIPAA) and growing volumes of image-based data such as scanned documents, IDs, screenshots, and forms, organizations face heightened risks of unintentional data exposure. The proposed solution leverages Optical Character Recognition (OCR) and machine learning–based entity detection to identify sensitive information within images and apply automated redaction. The assessment concludes that the solution is technically feasible, scalable, and capable of significantly reducing compliance and data privacy risks when implemented with appropriate controls.

2. Problem Statement

Organizations increasingly store and process images that contain sensitive personal information such as names, addresses, phone numbers, identification numbers, and financial details. Manual review and redaction of such images is time-consuming, error-prone, and not scalable. The absence of an automated PII redaction mechanism exposes organizations to regulatory non-compliance, reputational damage, and potential data breaches.

3. Objective

The primary objectives of this initiative are:

- To automatically detect PII present in images.
- To accurately redact or mask sensitive regions within images.
- To minimize manual intervention while maintaining high accuracy.
- To ensure compliance with data protection and privacy regulations.
- To provide a reusable and scalable framework that can be integrated into existing data pipelines.

4. Theory

The solution is based on a combination of computer vision and natural language processing techniques:

1. Optical Character Recognition (OCR): Converts text embedded in images into machine-readable text.
2. PII Detection: Applies rule-based patterns (e.g., regex for phone numbers, IDs) and/or trained NLP models (NER – Named Entity Recognition) to identify sensitive entities in extracted text.
3. Bounding Box Mapping: Links detected PII entities back to their spatial coordinates in the original image.
4. Redaction Techniques: Masks or blurs identified regions to permanently eliminate sensitive information from the image while preserving non-sensitive content.

5.Implementation

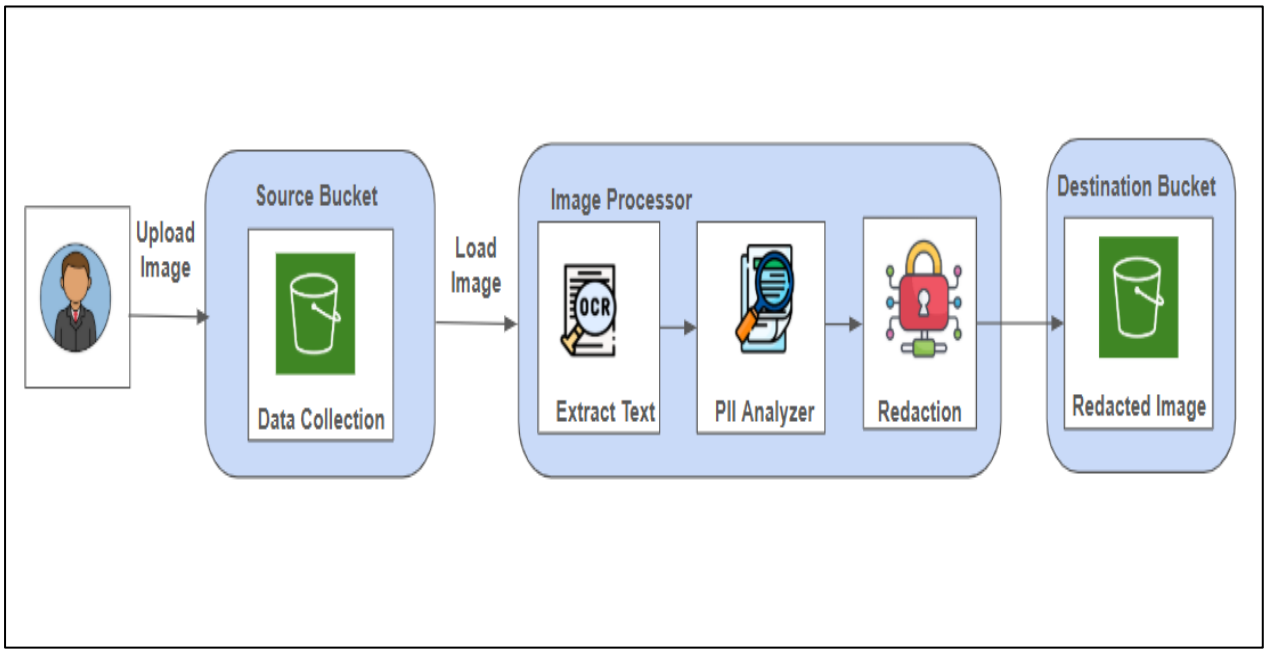

Fig 5 a: High Level Flow

The high-level implementation flow includes:

1. Image Ingestion: Input images are loaded from local storage or cloud repositories.
2. Text Extraction: OCR engine extracts text along with bounding box coordinates.
3. PII Identification: Detected text is analysed using predefined rules and/or ML models to classify PII.
4. Redaction: Identified regions are redacted using masking or pixelation techniques.
5. Output Generation: Sanitized images are stored or forwarded to downstream systems.

The solution can be implemented as a standalone service or integrated into batch or real-time data pipelines.

6.Methodology

a. Method I: PII Removal Using OCR, Presidio Detection, and Image Inpainting

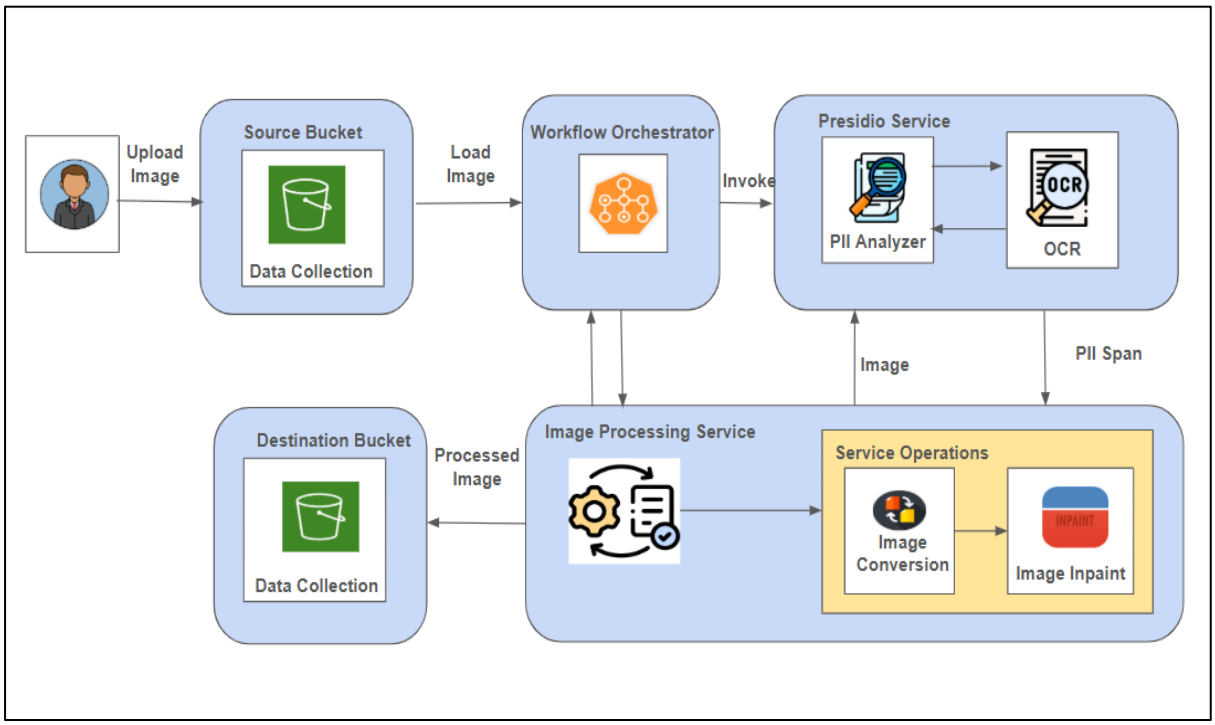

Fig 6 a: PII removal using Image Inpainting

1. Load the image from Amazon S3.
2. Detect Personally Identifiable Information (PII) using the Presidio analyser.

3. Convert the image from PIL format to OpenCV format for processing.
4. Generate masks corresponding to the detected PII regions.
5. Remove the PII text by applying image inpainting on the masked regions.
6. Convert the processed image back to PIL format.
7. Consolidate and post-process results obtained from the OCR engine and Presidio service.
8. Upload the PII-eliminated image and the corresponding analysis results back to Amazon S3.

b. Method II: PII Removal Using Presidio Image Redactor and Analyzer

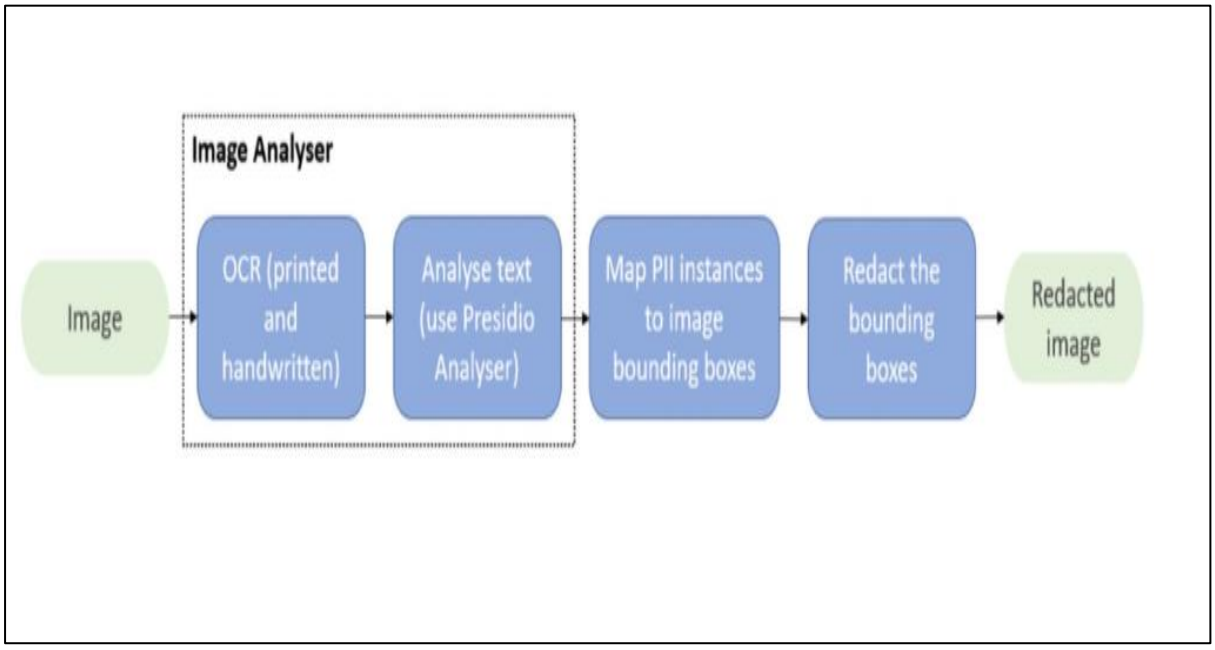

Fig 6 b: PII Redaction using Presidio

1. Load the image from Amazon S3.
2. Submit the image to the Presidio PII Redactor along with the configured PII recognizers.
3. Submit the image to the Presidio PII Analyzer along with the same PII recognizers.
4. Generate the redacted image directly using the Presidio Image Redactor.

5. Consolidate and post-process results obtained from the OCR engine and Presidio service.
6. Upload the redacted image and the corresponding analysis results back to Amazon S3.

7.Risk Mitigation

Risk	Description	Mitigation Strategy
OCR Accuracy	Poor image quality may lead to missed or incorrect text extraction	Image preprocessing, high-quality OCR engines, confidence thresholds
False Positives	Non-PII content may be incorrectly redacted	Fine-tuning rules/models, human review for critical workflows
False Negatives	Some PII may not be detected	Hybrid rule and ML approach, continuous model improvement
Performance	High processing time for large volumes	Parallel processing, GPU acceleration, batch optimization
Regulatory Misalignment	Changing compliance requirements	Configurable PII definitions and policies

8.Feasibility

a. Technical Feasibility

The solution is technically feasible using a robust and open-source–driven technology stack comprising PyTesseract, Microsoft Presidio, PIL (Python Imaging Library), and Pandas. PyTesseract enables reliable OCR for extracting text and bounding box coordinates from images, while PIL supports image preprocessing (resizing, denoising, grayscale conversion) and precise redaction through masking or pixelation. Microsoft Presidio provides a mature and extensible framework for PII detection using both predefined recognizers and customizable rules, ensuring flexibility across multiple PII types. Pandas is used to generate structured analysis and audit reports, enabling traceability of detected PII entities, confidence scores,

and redaction actions. All components are Python-native, well-documented, and widely adopted, making integration, maintenance, and future enhancements straightforward.

b. Operational Feasibility

Operationally, the solution integrates seamlessly into existing data processing workflows. The use of Python-based libraries allows easy deployment as a batch job, API service, or pipeline component. Automated PII detection and redaction significantly reduce manual effort and turnaround time for image sanitization.

c. Economic Feasibility

From an economic standpoint, the solution is highly cost-effective. The core technologies—PyTesseract, Presidio, PIL, and Pandas—are open source, minimizing licensing costs. Infrastructure expenses are limited to compute and storage resources required for OCR and image processing, which can be scaled based on workload.

d. Cost Analysis

The cost analysis outlines the estimated infrastructure and operational expenses for the image processing and OCR solution based on defined assumptions and baseline usage.

The base infrastructure cost (capital/recurring services) amounts to $122.996, covering key components such as Secret Manager, data transfer, container registry, object storage, VPC, and OCR processing (PyTesseract). These costs represent the foundational setup required to run the solution reliably and securely.

For operational scaling, the analysis assumes:

- Average image size: 5 MB
- Images processed per user: 1 image
- Average processing time per image: ~196 seconds

Based on these assumptions, the total estimated cost comes to $123.95, which includes both infrastructure and processing overheads. The cost for image of different size is summarized below.

Number Of Images:1		
Size Of Image(MB)	**Approx Runtime (seconds)**	**Price($)**
5	196	126.087
10	392	124.914
25	980	127.790
50	1960	132.585
100	3920	142.173
500	19600	218.883
1000	39200	314.769

9.Limitations

- OCR performance is dependent on image quality, orientation, and font clarity.
- Handwritten text detection may be less accurate than printed text.
- Contextual PII (e.g., indirect identifiers) may be difficult to detect reliably.
- Complete automation without human validation may not be suitable for high-risk use cases.

10.Conclusion

The feasibility assessment confirms that automated PII detection and redaction in images is both technically and operationally viable. The approach significantly reduces manual effort, enhances data privacy compliance, and mitigates the risk of sensitive data exposure. While limitations exist, particularly around OCR accuracy and contextual understanding, these can be managed through hybrid approaches and continuous improvement.

11.Next Steps

1. Conduct a pilot implementation on a representative dataset.
2. Evaluate OCR and PII detection accuracy using defined metrics.

3. Fine-tune detection rules and models based on pilot results.
4. Integrate the solution with existing data ingestion and storage systems.
5. Establish governance, audit, and human-in-the-loop review mechanisms.
6. Plan for scalability and future enhancements such as handwriting recognition and multilingual support.

Appendix

a. Glossary

i. **OCR (Optical Character Recognition)**

A technology used to extract machine-readable text from images, scanned documents, or PDFs.

ii. **PII Data (Personally Identifiable Information)**

Any information that can be used to identify an individual, either directly or indirectly, such as names, addresses, phone numbers, email addresses, government-issued IDs, or biometric data.

iii. **Inpainting**

An image-processing technique used to reconstruct or fill in selected regions of an image, typically to remove sensitive or unwanted content while preserving visual continuity.

iv. **Redaction**

The process of permanently removing or obscuring sensitive information from documents or images to prevent unauthorized access or disclosure.

v. **Orchestration Service**

A service responsible for coordinating and managing the execution flow of multiple components or services in a system, ensuring proper sequencing, data exchange, and error handling.

vi. **Pandas**

A Python data analysis and manipulation library that provides data structures and functions for handling structured data, such as tables and time-series data.

vii. **Presidio**

An open-source data protection framework developed by Microsoft that provides tools for detecting, analysing, and anonymizing Personally Identifiable Information (PII) in text and images

b. Screenshots

The figure below shows the pii text redacted using OpenCV in paint technique with aid of presidio analyser.

Fig 12 a: PII Redaction using Presidio for Sample 1

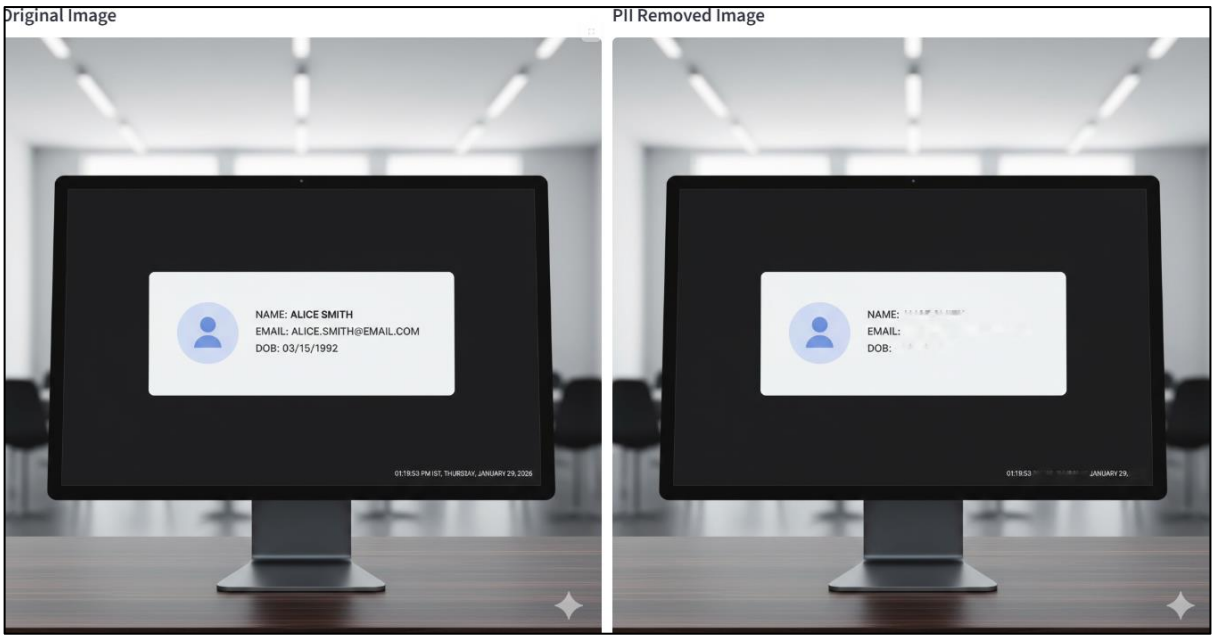

Fig 12 b: PII Redaction using Presidio for Sample 2